Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2025-P2-Model)** **Date: 13-05-2026**

Time: 02.30Pm to 05.30Pm **GTA-32** **Max. Marks: 180**

KEY SHEET

MATHEMATICS

1	**C**	2	**B**	3	**C**	4	**C**	5	**AB**	6	**BCD**
7	**ABD**	8	**AC**	9	**8**	10	**7**	11	**2**	12	**5**
13	**12**	14	**0.5**	15	**0**	16	**2025**				

PHYSICS

17	**C**	18	**B**	19	**A**	20	**A**	21	**AD**	22	**ABCD**
23	**ACD**	24	**ABC**	25	**3**	26	**5**	27	**3**	28	**4**
29	**19**	30	**2**	31	**1.75**	32	**8**				

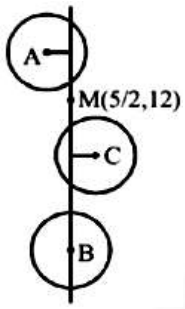
CHEMISTRY

33	**A**	34	**B**	35	**A**	36	**B**	37	**CD**	38	**ABCD**
39	**AD**	40	**AB**	41	**112**	42	**50**	43	**–1.46**	44	**36**
45	**0.8**	46	**140**	47	**32**	48	**4**				

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SOLUTIONS

MATHEMATICS

1. 

A(0, 16), B(3, 0), C(5, 8)

Line will pass through the mid point of A and C

$$m_{AB}=\frac{12}{-1/2}=-24\left|\frac{m}{8}\right|=3$$

2. $\left(1+x+2x^2+4x^4\right)20$

$$T_{r+1}={}^{20}C_r\cdot\left(1+x+2x^2\right)^{20-r}\cdot\left(4x^4\right)^r$$

For $r=19$

$$T_{19+1}={}^{20}C_{19}\cdot\left(1+x+2x^2\right)^1\cdot\left(4x^4\right)^{19}$$

$\therefore$ Coefficient of $x^{78}=2\cdot{}^{20}C_{19}\cdot4^{19}=2\cdot20\cdot2^{38}=10\cdot2^{40}$

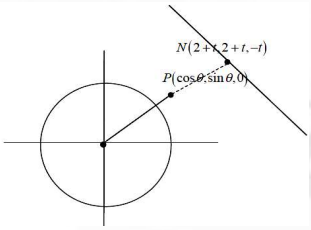
3. $N(2+t_1\,2+t_1-t)$

Dr's of PN are $(2+t-\cos\theta,2+t-\sin\theta-t)$

PN is perpendicular to the line $(2+t-\cos\theta,2+t-\sin\theta,-t).(1,1,-1)=0$

$2+t-\cos\theta+2+t-\sin\theta+t=0$ $\quad 3t+4=\cos\theta+\sin\theta$

$$t=\frac{\cos\theta+\sin\theta-4}{3}$$



$$PN=\sqrt{(2+t-\cos\theta)^2+(2+t-\sin\theta)^2+t^2}$$

$$\frac{\sqrt{10-(4\sin\theta+4\cos\theta+2\sin\theta\cos\theta)}}{\sqrt{3}}$$

$$(PN)=(PN)_{nm}=\frac{\sqrt{9-4\sqrt{2}}}{3}@\theta=\frac{\pi}{4}$$

4. Observe that 1, 2, 3, 4 are roots of $n^4-wn^3-zn^2-yn-x=0$

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$w=10, z=-35, y=50, x=-24$

5. $\sin 6\beta+\sin 2\beta=3\cos 4\beta$

$$\Rightarrow 2\sin 4\beta\cdot\cos 2\beta=3\cos 4\beta \Rightarrow \frac{\sin 4\beta\cdot\cos 2\beta}{\cos 4\beta}=\frac{3}{2}$$

$$\Rightarrow \frac{\sin 2\beta\cos^2 2\beta}{\cos 4\beta}=\frac{3}{4} \Rightarrow \frac{\sin(\alpha-3\beta)\cdot\cos^2 2\beta}{\cos^2 2\beta-\sin^2 2\beta}=\frac{3}{4}$$

$$\Rightarrow \frac{2}{\sin\left(\frac{4\beta}{4}\right)}=\frac{1}{\cos\alpha}+\frac{1}{\sin\beta} \Rightarrow \frac{1}{\sin\beta}=\frac{1}{\cos\alpha} \quad \therefore \sin\beta=\cos\alpha \Rightarrow \cos\left(\frac{\pi}{2}-\beta\right)=\cos(5\beta)$$

$$\Rightarrow 5\beta=2n\pi\pm\left(\frac{\pi}{2}-\beta\right) \quad \therefore \beta=(4n-1)\frac{\pi}{8} \text{ or } \beta=(4n+1)\frac{\pi}{12}$$

6. $P(X)=P(X_3)+P(\overline{X_3}\cap X_1\cap X_2) \quad =\frac{3}{4}+\frac{1}{4}\cdot\frac{8}{10}\cdot\frac{9}{10}=\frac{93}{100}$

$$P\left(\frac{\textit{Exactly two working}}{X}\right)\neq 1 \qquad P\left(\frac{X}{X_2}\right)=\frac{\frac{9}{10}\cdot\frac{8}{10}+\frac{9}{10}\cdot\frac{2}{10}\cdot\frac{3}{4}}{\frac{9}{10}}=\frac{19}{20}$$

7. (A) $\sum\frac{\sqrt{\sin A}}{\sqrt{\sin B}+\sqrt{\sin C}-\sqrt{\sin A}}a=k\sin A$

$\sum\frac{\sqrt{a}}{\sqrt{b}+\sqrt{c}-\sqrt{a}}$ Let $\sqrt{b}+\sqrt{c}-\sqrt{a}=x$

$\sqrt{c}+\sqrt{a}-\sqrt{b}=y \quad \sum\frac{y+z}{2x}$ $\sqrt{a}+\sqrt{b}-\sqrt{c}=z$

$\frac{1}{2}\sum\frac{y+z}{x}$ $\sqrt{a}+\sqrt{b}+\sqrt{c}=x+y+z$

$\frac{1}{2}\sum\left(\frac{y}{x}+\frac{z}{x}+\frac{x}{y}+\frac{z}{y}+\frac{y}{z}+\frac{x}{z}\right)$ $\frac{1}{2}\left(\frac{y}{x}+\frac{z}{x}+\frac{x}{y}+\frac{z}{y}+\frac{y}{z}+\frac{x}{z}\right)$

$\frac{1}{2}\left(\frac{y}{x}+\frac{x}{y}\right)+\left(\frac{x}{z}+\frac{z}{x}\right)+\left(\frac{y}{z}+\frac{z}{y}\right)$ $\frac{1}{2}(\geq(2+2+2)=\geq 3)$

(B, C) $(\sec A\sec B+\tan A\tan B)^2-(\sec A\tan B+\tan A\sec B)^2=1$

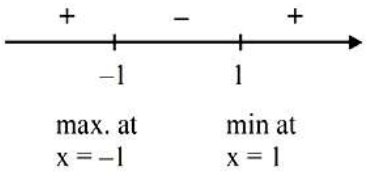
(D) $U(0,1,0),\ M(-1,0,0),\ R(1,0,0)$ orthocenter is $(0,1,0)$

8. (A) $f(x)=\sin^{-1}\left(\log_3\left(\frac{x^2-x+1}{x^2+x+1}\right)\right)$

$$f(-x)=\sin^{-1}\left(\log_3\left(\frac{x^2+x+1}{x^2-x+1}\right)\right)=-f(x)$$

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$$f\left(\frac{1}{x}\right)=f(x)=-f(-x)$$

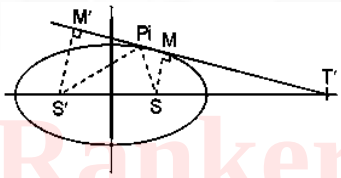


(B) $f'(x)=\dfrac{1}{\sqrt{1-\left(\ell og_3\left(\dfrac{x^2-x+1}{x^2+x+1}\right)\right)^2}}\cdot\dfrac{\ell og_3 e}{\left(\dfrac{x^2-x+1}{x^2+x+1}\right)}\cdot\dfrac{2\left(x^2-1\right)}{\left(x^2+x+1\right)^2}$

Since $\dfrac{x^2-x+1}{x^2+x+1}\in\left[\dfrac{1}{3},3\right]$ $\quad\therefore f(x)=\left[\dfrac{-\pi}{2},\dfrac{\pi}{2}\right]$

9. Area of $\Delta P_1ST_1=\dfrac{1}{2}P_1T_1\times SM$, Area of $\Delta P_1T_1S'=\dfrac{1}{2}P_1T_1\times S'M'$

$$\frac{Area\left(\Delta P_1T_1S\right)\cdot Area\left(\Delta P_1T_1S'\right)}{\left(P_1T_1\right)^2}=\frac{1}{4}SM\cdot S'M'=\frac{b^2}{4}$$



$$\sum_{i=1}^{n}\frac{Area\left(\Delta PiTiS\right)\cdot Area\left(\Delta PiTiS'\right)}{\left(PiTi\right)^2}=\frac{1}{4}SM\cdot S'M'=\frac{b^2}{4}$$

$$\sum_{i=1}^{n}\frac{Area\left(\Delta PiTiS\right)\cdot Area\left(\Delta PiTiS'\right)}{\left(PiTi\right)^2}=\frac{nb^2}{4}$$

$$\frac{nb^2}{4}=18\Rightarrow n=\frac{4\times18}{9}=8\Rightarrow n=8$$

10. We have $f'(x)=f'(3-x)$ on integrating w.r.t. x

$f(x)=-f(3-x)+c$

Put $x=0$ $\quad f(0)+f(3)=c$ $\quad -32+46=c$ $\quad c=14$

Now let $I=\int_0^3 f(x)dx$ $\quad\ldots\ldots\ldots(1)$

$I=\int_0^3 f(3-x)dx$ $\quad\ldots\ldots\ldots(2)$

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Adding (1) and (2), we get $2I=\int_0^3\left(f(x)+f(3-x)\right)dx=\int_0^3 14dx=42$

11. $B^2-tr(B)\cdot B+I=O \Rightarrow AB-\left(tr(B)\right)A+AB^{-1}=O$

$\therefore tr(AB)-tr(A)tr(B)+tr\left(AB^{-1}\right)=O$

12. Let $a+2b+c=x, a+b+2c=y, a+b+3c=z$

$a=-x+5y-3z, b=z+x-2y, c=z-y$

Apply $A.M.\geq G.M.$ $\frac{a+3c}{a+2b+c}+\frac{4b}{a+b+2c}-\frac{8c}{a+b+3c}\geq -17+12\sqrt{2}$

13. $\Delta=\frac{1}{2}\cdot 8\cdot P=4P \quad P=\left|\frac{3\cos\theta-4-\sin\theta}{\sqrt{2}}\right|$

$2\sqrt{2}\left(4-\sqrt{10}\right)=\Delta_1 \quad 2\sqrt{2}\left(4+\sqrt{10}\right)=\Delta_2$

14. Let $t=\tan x \qquad x\to\frac{\pi^-}{2},\ t\to\infty$

$\lim_{t\to\infty}(1+t)\left\{(1+t)\ell n\left(\frac{1+t}{2+t}\right)+1\right\}$

$\Rightarrow \lim_{t\to\infty}(1+t)\left\{(1+t)\ell n\left(1-\frac{1}{2+t}\right)+1\right\}=\lim_{t\to\infty}(1+t)\left\{(1+t)\left\{-\frac{1}{2+t}-\frac{1}{\frac{(2+t)^2}{2}}+.....\right\}+1\right\}$

$=\lim_{t\to\infty}(1+t)\left\{\left\{-\frac{1+t}{2+t}-\frac{(1+t)^2}{(2+t)^2}+....\right\}+1\right\}=\lim_{t\to\infty}(1+t)\left\{\left(-1+\frac{1}{2+t}-\frac{2(1+t)}{(2+t)^2}.....\right)+1\right\}$

$=1-2=-1$

15. $g(x)=\frac{f(x)-f(-x)}{|x|+2}=\frac{8x^3-4x}{|x|+2} \qquad \therefore g(x)=g(-x) \quad \therefore g(x)$ is odd.

16. Since $f(x)=\left(1+x+x^2+.....+x^n\right)^2$

$f'(x)=2\left(1+2x+.....+nx^{n-1}\right)\left(1+x+x^2+.....+x^n\right)$

So, $\begin{cases} C(r,n)=(r+1)(r+2)\ if\ 0\leq r\leq n-1, r\in Z \\ C(r,n)=(r+1)(2n-r)\ if\ n\leq r\leq 2n-1, r\in Z \end{cases}$

This can be easily checked with induction or otherwise.

Particularly in our case pattern helps as well: $f(x)=\left(1+x+.....+x^{2026}\right)^2$

$f'(x)=2\left(1+x+x^2+.....+x^{2026}\right)\left(1+2x+3x^2+.....+2026x^{2025}\right)$

$C(1,2026)=2.3, \quad C(2,2026)=3.4$

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PHYSICS

17. $\oint \overline{B}.\overline{dl} = \mu_0 I\left(\frac{\Omega}{4\pi}\right) = \mu_0 I\left(1-\frac{\sqrt{3}}{2}\right)$

18. Mirror has rotate 45°.

19. $F = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left\{\frac{Iahr\cos\theta}{c} + \frac{2Iphr\cos^3\theta}{c}\right\} d\theta$

$= \frac{38}{15}\frac{Irh}{c}$

20. $P_{below} = P_{air} - \frac{4T\cos\theta}{d}$

$\Rightarrow P\left(\frac{l}{l-h}\right) - \frac{4T\cos\theta}{d} = P_0 - \rho gh$

$\Rightarrow T = \frac{d}{4\cos\theta}\left\{\frac{P_0 h}{l-h} + \rho gh\right\}$

21. $i = \int_0^l \left(ae^{bx}\right).l\,dx = al\left[\frac{e^{bx}}{b}\right]_0^l = \frac{al}{b}\left(e^{bl}-1\right)$

22. Net magnetic field at O is zero

23. Time constant of RL circuit is $\frac{5}{3}\Omega$

24. $v = \sqrt{\frac{T}{\mu}} = 50\,m/s$

$P_{av} = \frac{1}{2}\mu v w^2 A^2 = 0.126\,w$

25. Apply principle of conservation of linear momentum and conservation of energy

26. $pv\omega^2 r\sin^2 \; pvg\cos\theta$

$\Rightarrow r = \frac{g\cos\theta}{\omega^2\sin^2\theta} = \frac{2\sqrt{3}\,g}{\omega^2}$

27. $P_2 = \rho gh - \frac{1}{2}\rho\left(v_2 - v_1{}^2\right)$ .....(1)

$P_2 + \frac{1}{2}(2\rho)v_2{}^2 + 2\rho gh = \frac{1}{2}(2\rho)v_2{}^2$ .....(2)

$4Av_1 = Av_2 = av_3$ .....(3)

$\Rightarrow v_3 = \sqrt{\frac{3gh}{1-\frac{17}{32}\frac{a^2}{A^2}}}$

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28. $U_{total}=\frac{Kq^2}{2a}-\frac{Kq^2}{b}+\frac{Kq^2}{2b}=\frac{Kq^2}{2a}-\frac{Kq^2}{2b}$

29. $U=\frac{\rho^2g^2l^2}{B}\int h^2dh=\frac{\rho^2g^2l^2}{B}\left(\frac{h^3}{3}\right)_{h_1}^{h_2}$

$\frac{U_{lower}}{U_{upper}}=19$

30. $\tau=\int_O^R\frac{2\mu mg}{R^2}r^2dr=\frac{2}{3}\mu mgR$

31. Kinetic energy will be maximum when hemisphere turns through $\frac{\pi}{2}$

COE; $\frac{1}{2}\left(\frac{2}{3}mR^2\right)\omega^2=mg\frac{R}{2}\Rightarrow\omega^2=\frac{3g}{2R}$

N2L; $N-mg=m\omega^2\frac{R}{2}$

$\Rightarrow N=\frac{7}{2}mg$

32. $(\lambda x)g=(\lambda L)a\Rightarrow a=\frac{x}{L}g$

$T=\lambda(L-x)\left(\frac{x}{L}-g\right)$

$at\ x=\frac{L}{2}$

$T_{\max}=\lambda\frac{1}{L}g$

For rope not to break

$\frac{\lambda Lg}{4}\le 2\lambda h\Rightarrow L\le 8m$

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CHEMISTRY

33. NO_2 in III is out of plane.

34. Neglecting the attraction term gives $P(V_m - b) = RT$. Thus $Z_c = PV_m / RT = V_m / (V_m - b) > 1$. The excluded volume reduces free space. Collision frequency is proportional to effective number density. Since molecules occupy free volume $(V_m - b)$, density increases compared to an ideal gas. Therefore, collisions are more frequent.

35. NCERT

36. Sodium cyanide selectively depresses ZnS by forming a stable soluble cyanide complex. Because this complex remains in solution, H_2S added directly cannot precipitate the metal sulphide. This indicates that (X) is a strongly bound zinc cyanide complex. Acidification destroys the cyanide complex, releasing Zn^{2+} ions. When H_2S is then introduced, zinc sulphide precipitates from solution, giving the observed solid (Y).

37. P is tertiary alcol, Q is acetophenone, R is ketone.

38. K decreases with temperature; hence the reaction is exothermic and enthalpy-driven. Hydrophobic ligand L causes ordering of water in the reactant state, giving $\Delta S^\circ < 0$ (A). Diamagnetic $\left[M(L)_6\right]^{2+}$ indicates a strong-field ligand, increasing CFSE and making ΔH° negative (C). Redefining the standard state of L to 1 M aqueous introduces translational entropy, increasing $\left|\Delta S^\circ\right|$ (D).

39. A) Copper is oxidized by oxygen in acidic medium to form $CuSO_4$.
B) Copper with dilute HNO_3 gives NO (not N_2O).
C) Excess CN^- reduces Cu^{2+} to Cu^+ forming cyanide complex.
D) NH_4OH precipitates $Cu(OH)_2$ from Cu^{2+} solution.

40. A primary amine gives positive test with Hinsberg reagent $(PhSO_2C\ell), a - NO_2$ groupgives positive test with Mulliken's Reagent $(Zn / NH_4C\ell)$, Tollen's reagent and an acidicfunctional group give positive test with sodium metal.

41. At pH = 6, lysine has an extra protonated amino group, giving it the highest positive charge → option 1.
Stability of group 14 hydrides decreases down the group, so SiH_4 is most stable → option 1.

42. Membrane (cell 1 → cell 2)
$P_2 - P_1 = RT[B]$
Membrane (cell 2 → cell 3)
This membrane blocks solute C.
$P_3 - P_2 = RT[C]$
$P_3 - P_1 = RT([B] + [C])$
$P_3 - P_1 = 50\,bar$

43. From hydrolysis relation: pH = 4.8 + 1/2(−log C)

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At C = 0.1 M $\Rightarrow -\log C = 1 \Rightarrow$ pH = 5.3

CsCl lattice: Z = 1

$a = 2(r_+ + r_-)/\sqrt{3} = 400\,pm = 4\times10^{-8}\,cm$

$V = a^3 = 64\times10^{24}\,cm^3$

ρ = M/(NA·V) = M/38.4

$\Rightarrow M/(10\rho) = 3.84$

Required value = 3.84 − 5.3 = −1.46

44. (i) $[Cr(EDTA)]^-$ → octahedral chelate → chiral → active.

(ii) $[Ru(en)_3]^{2+}$ → tris-chelate → chiral → active.

(iii) $[Pt(gly)Cl_2]^-$ → square planar, unsymmetrical chelation → inactive.

(iv) $cis-[Cr(Cl)_2(ox)_2]^{3-}$ → cis-chelate octahedral → chiral → active.

(v) $trans-[Pt(en)(NH_3)_2(py)_2]^{4+}$ → chelate + mixed ligands remove symmetry → inactive.

45. $[S^{2-}] = K_{a1}.K_{a2}.\dfrac{[H_2S]}{[H^+]^2} = 10^{-8}\times10^{-13}\times\dfrac{0.1}{(10^{-3})} = 10^{-16}$ M

46. Cyclohexene on oxidative cleavage with hot acidic $K_2Cr_2O_7$ gives adipic acid. Heating with $Ca(OH)_2$ causes ketonic decarboxylation to form cyclopentanone (P).
Cyclopentanone forms an oxime which undergoes Beckmann rearrangement to a lactam. Reduction with LAH gives a cyclic amine (Q). Another portion of cyclopentanone forms a cyanohydrin which on LAH reduction followed by HNO_2 treatment gives a diol (R). Condensation of cyclic amine (Q) with diol (R), followed by acetylation–hydrolysis, produces T having molecular formula $C_7H_{12}O_2$.
Molar mass = $(7\times12)+(12\times1)+(2\times16) = 140$.

47. $\Delta E_{21} = 13.6Z^2(1-1/4) = 13.6Z^2\times3/4$

$\Delta E_{31} = 13.6Z^2(1-1/9) = 13.6Z^2\times8/9$

$P'_{ph}/P_{ph} = \Delta E_{31}/\Delta E_{21} = 32/27$

48. In ester from carboxylic part as well as alcohol part are considered.

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